

A countability obstruction to a full interval of graph-invariant gaps

A disproof of one assertion of conjecture 00000003480

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Conjecture 00000003480 states that the difference between the fractional chromatic number and a fractional Hall bound takes every real value from 0 to 1, with a Mycielski-type family realizing the values. For finite unweighted simple graphs, such a full interval is impossible for any pair of real-valued graph invariants.

For each $n \geq 0$ there are at most $2^{\binom{n}{2}}$ simple graphs on a labeled n -element vertex set. Hence the disjoint union of the graph sets over all n is countable. Every finite simple graph is isomorphic to one in this union. If A and B are any real-valued graph invariants, their difference has at most countably many values:

$$\{A(G) - B(G) : G \text{ a finite simple graph}\} \text{ is countable.}$$

In contrast, $[0, 1]$ is uncountable. For example, the injective map

$$x \mapsto \frac{\arctan(x) + \pi/2}{\pi}$$

sends $\mathbb{R}$ into $(0, 1)$. Therefore

$$[0, 1] \not\subseteq \{\chi_f(G) - H_f(G) : G \text{ a finite simple graph}\},$$

where χ_f and H_f denote the two invariants in the conjecture. The asserted continuous spectrum fails, regardless of the definitions of these invariants or the behavior of the endpoint 1.

Formalization. The Lean file encodes the class of all finite labeled simple graphs as $\Sigma n : \mathbb{N}, \text{SimpleGraph (Fin } n)$. It proves this class countable and the real interval $[0, 1]$ uncountable. The theorem `TLMC3480.no_full_interval_of_difference` states that the difference of any two real-valued functions on this graph class cannot take every value in $[0, 1]$. The arbitrary-function statement applies to the named fractional graph invariants without using their LP formulas. Its axiom audit contains only `propext`, `Classical.choice`, and `Quot.sound`.

Scope. The argument addresses finite unweighted simple graphs, the standard domain of the described Mycielski construction. A differently stated conjecture allowing continuously weighted or otherwise uncountably parametrized graph objects is outside its scope.